

Quantum-Assisted Multi-Asset Portfolio Construction

Step 07

Independent Classical Validation

Reconstructing the optimization with CVXPY, KKT certificates, and exact enumeration

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1 Why independent validation is necessary

An optimizer should not certify itself. The original continuous solutions were produced by SciPy SLSQP. Step 7 reconstructs the mathematical problem independently in CVXPY and uses Clarabel or OSQP as the reported validation solver. The stored portfolio is only an initial point; it is prohibited from becoming the validation answer without a new solve.

2 Independent quadratic-program form

The lifted variable is $x = (w, p, n, h)$. The independent problem is written as

$$\min_x \frac{1}{2}x^\top Hx + f^\top x + c \quad (1)$$

subject to

$$A_{\text{eq}}x = b, \quad A_{\text{ineq}}x \leq d, \quad \ell \leq x \leq u. \quad (2)$$

The matrix and vectors are rebuilt from the Step 3 data and Step 5 policy context rather than copied from the original solver's internal state.

3 Convexity certificate

A quadratic program is convex when its Hessian satisfies

$$H = H^\top, \quad \lambda_{\min}(H) \geq -\varepsilon. \quad (3)$$

Step 7 checks symmetry and the minimum eigenvalue. Convexity means any KKT point satisfying regularity conditions is a global optimum, not merely a local solution.

4 KKT derivation

For equality constraints $Ax = b$, inequality constraints $Gx \leq h$, and bound constraints, define Lagrange multipliers ν , $\lambda \geq 0$, $\alpha \geq 0$, and $\beta \geq 0$. The Lagrangian is

$$\mathcal{L}(x, \nu, \lambda, \alpha, \beta) = \frac{1}{2}x^\top Hx + f^\top x + \nu^\top (Ax - b) + \lambda^\top (Gx - h) + \alpha^\top (\ell - x) + \beta^\top (x - u). \quad (4)$$

The KKT conditions are:

$$\text{Primal feasibility: } Ax = b, \quad Gx \leq h, \quad \ell \leq x \leq u, \quad (5)$$

$$\text{Dual feasibility: } \lambda, \alpha, \beta \geq 0, \quad (6)$$

$$\text{Stationarity: } Hx + f + A^\top \nu + G^\top \lambda - \alpha + \beta = 0, \quad (7)$$

$$\text{Complementarity: } \lambda_j (Gx - h)_j = 0, \quad \alpha_i (\ell_i - x_i) = 0, \quad \beta_i (x_i - u_i) = 0. \quad (8)$$

Step 7 reports infinity norms of primal, stationarity, and complementarity residuals and the minimum dual value.

5 Solver independence and diagnostics

CVXPY Clarabel and OSQP are the eligible independent solvers. SciPy trust-constr and a split-constraint SLSQP run are retained only as diagnostics. This hierarchy avoids labeling a second call to the original algorithm as independent validation.

6 Fixed-support validation

For the QAOA support, inactive assets are constrained to remain at incumbent weights. The independent QP then solves exact continuous weights over the active support. The same process is applied to the exact enumerated support. Because the supports are identical in the final run, the independently refined portfolios are identical.

7 Numerical results

Validation target	Selected solver	Constraints	Failures	Max violation	Weight L1 difference
Primary global	CVXPY Clarabel	130	0	2.22e-16	2.07e-10
Strict global	CVXPY Clarabel	130	0	7.77e-16	9.08e-07
QAOA fixed support	CVXPY OSQP	174	0	1.04e-17	1.57e-14
Exact fixed support	CVXPY OSQP	174	0	1.04e-17	1.57e-14

All hard-constraint audits pass. The reported weight differences between stored and independently solved portfolios are at numerical precision. All KKT certificates pass.

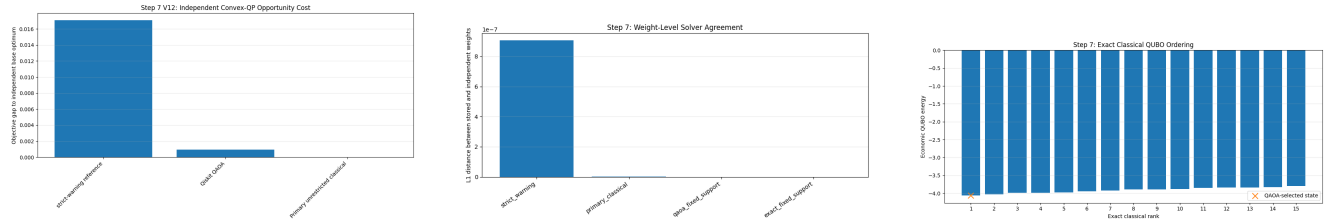


Figure 1: Independent validation diagnostics exported by Step 7.

8 Exact binary enumeration

For each six-element subset of the 11 candidates, Step 7 evaluates the exact QUBO energy. The number of states is 462. The selected QAOA support has classical rank 1 and zero energy gap:

$$E_{\text{QAOA}} - E_{\text{exact}} = 0. \quad (9)$$

This is the binary ground truth for the reduced problem.

9 Objective reconciliation

The project objective is decomposed into growth reward, income reward, variance penalty, concentration penalty, linear/turnover cost, impact cost, and scenario penalty. Step 7 recomputes those components from the independent weights and verifies that their sum equals the reported objective within tolerance.

10 Formal verdict

The formal verdict is `PASS_QAOA_EXACT`. It means:

- the selected best-of-20 QAOA support matches the exact reduced QUBO optimum;
- the support-constrained continuous portfolio is independently reproduced;
- all applicable hard constraints pass; and
- KKT and convexity checks pass.

`PASS_QAOA_EXACT` denotes agreement with the exact reduced-QUBO ground state plus successful fixed-support continuous reconstruction. It does not claim exhaustive exact-finance optimality across all 462 supports or quantum advantage.

11 Forward-path validation

The three certified shortlist portfolios are evaluated on 1,000 common heavy-tailed synthetic paths. This is an additional model robustness test, not an independent historical sample. The raw path-level matrix was not exported in the final package, so bootstrap winner confidence intervals are not claimed.

12 Reproduction checklist

Rebuild the QP independently; use a different solver family; prohibit stored weights from being selectable; verify Hessian PSD; export primal, stationarity, complementarity, and dual residuals; enumerate all binary states; reoptimize QAOA and exact supports; and reconcile every objective component.

13 What Step 7 establishes

Step 7 provides strong evidence that the code solved the declared synthetic optimization problems correctly. It does not validate the economic truth of expected returns or guarantee out-of-sample performance. Solver correctness and market validity are different questions.

A Building an independent validation contract

Here independent means independent solver/reconstruction under the same declared financial model: SLSQP generates the original solution, while CVXPY with Clarabel or OSQP rebuilds and solves the QP. This checks solver/self-validation errors; shared input assumptions remain common.

The contract records the original solver, eligible validator solvers, numerical tolerances, variable ordering, objective convention, policy context, and selected support. A change to any of these invalidates a direct comparison unless the contract is regenerated.

B Lifted QP reconstruction

Let

$$x = \begin{bmatrix} w \\ p \\ n \\ h \end{bmatrix}. \quad (10)$$

The independent program is

$$\min_x \frac{1}{2} x^\top H x + f^\top x + c \quad (11)$$

subject to

$$A_{meq}x = b, \quad A_{mineq}x \leq d, \quad \ell \leq x \leq u. \quad (12)$$

Every block is reconstructed from saved economic inputs. The validator does not copy a numerical Hessian or constraint matrix from SLSQP memory. This protects against a shared implementation error in matrix assembly.

C Convexity certificate in detail

The validator checks

$$\|H - H^\top\|_\infty \leq \epsilon_{\text{sym}}, \quad (13)$$

$$\lambda_{\min}(H) \geq -\epsilon_{\text{psd}}. \quad (14)$$

A tiny negative eigenvalue can arise from floating-point arithmetic, so the tolerance is explicit. If the Hessian is materially indefinite, KKT satisfaction would no longer be sufficient for global optimality and the reported convex validation would fail.

D KKT residual definitions

For equality residual $r_{\text{eq}} = A_{meq}x - b$ and inequality slack $s = d - A_{mineq}x$, primal residual can be summarized by

$$r_{\text{primal}} = \max \left(\|r_{\text{eq}}\|_\infty, \max_j [-s_j]_+, \max_i [\ell_i - x_i]_+, \max_i [x_i - u_i]_+ \right). \quad (15)$$

Stationarity residual is

$$r_{\text{stat}} = \|Hx + f + A_{\text{eq}}^\top \nu + A_{\text{ineq}}^\top \lambda - \alpha + \beta\|_\infty. \quad (16)$$

Complementarity residual is the largest magnitude among

$$\lambda_j (A_{\text{ineq}}x - d)_j, \quad \alpha_i (\ell_i - x_i), \quad \beta_i (x_i - u_i). \quad (17)$$

Dual feasibility checks the minimum multiplier. A certificate passes only when all required residuals are below declared tolerances.

E Why objective agreement alone is insufficient

Two portfolios can have nearly equal objective values while one violates a constraint or has materially different weights. Step 7 therefore checks objective gap, weight L^1 distance, maximum weight difference, full constraint rows, and KKT residuals. Conversely, a larger weight difference can still be legitimate if the objective is flat and both solutions are optimal; the complete evidence must be read together.

F Global and fixed-support problems

The primary and strict-warning portfolios are validated as global continuous problems. The QAOA and exact active-set portfolios are validated with inactive assets fixed to incumbent values. If $\mathcal{A}$ is active support, then

$$w_i = w_i^0 \quad (i \notin \mathcal{A}). \quad (18)$$

This creates extra equality constraints, explaining why the fixed-support audit has more rows than the global audit.

G Exact enumeration algorithm

Enumerate every subset $\mathcal{A} \subset \{1, \dots, 11\}$ with $|\mathcal{A}| = 6$. For each support:

1. construct its bitstring;
2. evaluate the QUBO energy under the stored convention;
3. optionally solve the exact fixed-support continuous problem;
4. store energy, rank, feasibility, and refined objective.

Sorting the 462 energies establishes exact QUBO rank. The selected QAOA state has rank one and gap zero.

H Objective-component reconciliation

The validator recomputes

$$F = F_{\text{growth}} + F_{\text{income}} + F_{\text{variance}} + F_{\text{concentration}} + F_{\text{linear}} + F_{\text{impact}} + F_{\text{scenario}}. \quad (19)$$

Each component is calculated from independent weights and primitive inputs. The sum must agree with the selected solver objective after accounting for any stored constant and sign convention. This catches missing factors of two, annualization mistakes, and inconsistent cost units.

I Interpreting the numerical evidence

Clarabel is selected for the primary and strict global problems, and OSQP for the support-constrained problems. The final audits report zero failures, maximum violations near machine precision, and very small stored-versus-independent weight differences. The strict portfolio has a larger but still negligible L^1

difference than primary because its solution lies near active warning constraints and solver tolerances can produce small reallocations.

All selected KKT certificates pass. Exact and QAOA supports produce identical independent refinements. These results justify the formal verdict `PASS_QAOA_EXACT` within the declared synthetic model.

J Forward-path evaluation

Common heavy-tailed synthetic paths are generated for the three certified candidates. Using the same path for every portfolio reduces comparison noise. For path m , cumulative wealth and drawdown are computed from portfolio returns. Summary statistics include median return, fifth-percentile return, median drawdown, 95th-percentile drawdown, and loss frequency.

This is model-based robustness evidence. It is not an independent historical backtest because the paths are generated from the same broad return-and-risk assumptions. The raw path matrix was not exported, so the package does not claim bootstrap confidence intervals.

K Failure conditions

Step 7 must fail if the Hessian is not PSD, no independent solver returns an acceptable optimum, a hard constraint fails, KKT residuals exceed tolerance, objective reconstruction disagrees, selected QAOA energy is above exact minimum beyond tolerance, or the reported support and decoded bitstring disagree.

L Reproduction protocol

1. Load only exported economic inputs and selected portfolios.
2. Rebuild H, f, A, b, d, ℓ, u independently.
3. Solve with at least one eligible CVXPY solver.
4. Produce a full constraint table from the independent solution.
5. Recover dual variables and calculate KKT residuals.
6. Reconcile objective components and weight differences.
7. Enumerate all 462 supports and verify QAOA rank and gap.
8. Reoptimize QAOA and exact supports under identical fixed-support constraints.
9. Generate common forward paths and export summaries with explicit synthetic scope.